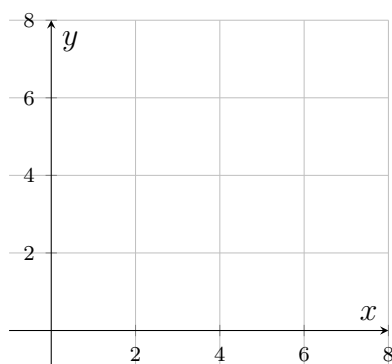


Quadrilaterals: Which Way Does the Implication Go?

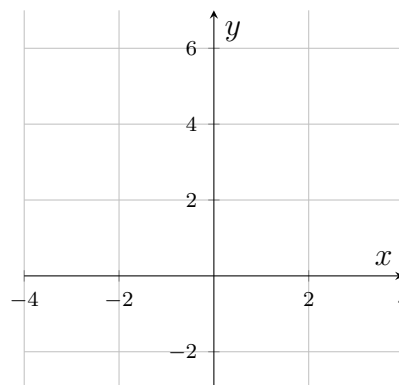
Skills practised: Testing one-way implications with algebra, side lengths, diagonals, and counterexamples.

Every square is a rhombus, but not every rhombus is a square: the implication runs one way only. Each problem below contains a claim that runs the implication backwards, or asks you to decide how far the evidence actually reaches. **Answer the computational part first, then write the diagnosis in a full sentence** — naming the property that was assumed and the property that was actually established.

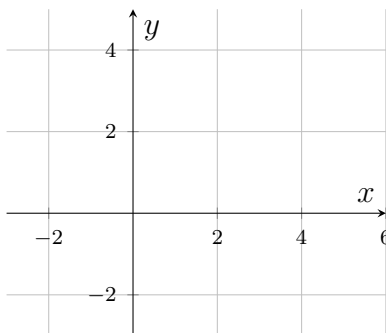
Plotting grids for problems 2, 4 and 6. The three coordinate problems each get a blank grid, printed together here and named. Every one of them says which grid it means — *no grid belongs to any other problem*. Plot on the grid, then work below the problem.



Grid 2 — for problem 2.



Grid 4 — for problem 4.



Grid 6 — for problem 6.

1. $MNPQ$ is a rhombus. One interior angle measures $(4x + 10)^\circ$ and a consecutive interior angle measures $(6x + 20)^\circ$.

(a) Find x . (b) Must $MNPQ$ be a square? Justify your answer using the angle measures you found.

Answer: _____

2. Quadrilateral $ABCD$ has vertices $A(1, 1)$, $B(5, 3)$, $C(7, 7)$ and $D(3, 5)$, and all four sides are congruent. Priya writes: “Four congruent sides, so $ABCD$ is a square — and therefore its diagonals are congruent, both about 8.49.”

(a) Compute the lengths of both diagonals AC and BD , to two decimal places. (b) Which of Priya’s two claims is the one that fails, and what is the most specific correct name for $ABCD$? (Plot $ABCD$ on Grid 2, printed with the directions.)

Answer: _____

3. For each statement, write **always true** or **not always true**. When it is not always true, describe one counterexample precisely enough that a classmate could draw it.

(a) If a quadrilateral is a rectangle, then it is a parallelogram.

(b) If a quadrilateral is a parallelogram, then it is a rectangle.

(c) If the diagonals of a quadrilateral are perpendicular, then it is a rhombus.

(d) If a quadrilateral is a square, then it is a rhombus.

4. Quadrilateral $ABCD$ has vertices $A(0, 6)$, $B(3, 0)$, $C(0, -2)$ and $D(-3, 0)$. Sam writes: *“One diagonal is vertical and the other is horizontal, so the diagonals are perpendicular — that makes $ABCD$ a rhombus, with every side about 6.71.”*

(a) Compute AB and BC , to two decimal places. (b) Sam’s diagonal observation is correct. Explain exactly what is missing from his argument, and name $ABCD$. (*Plot $ABCD$ on Grid 4, printed with the directions.*)

Answer: _____

5. $PQRS$ is a parallelogram whose diagonals measure $PR = 3x - 5$ and $QS = x + 11$. Sam sets the diagonals equal, writes $x = 3$, and concludes: “congruent diagonals, so $PQRS$ must be a square.”

(a) Find and fix Sam’s arithmetic slip: what is the correct value of x ? (b) With the corrected value, what is the most specific name $PQRS$ is guaranteed to have — and what extra piece of information would be needed to reach “square”?

Answer: _____

6. Challenge. Quadrilateral $WXYZ$ has vertices $W(-1, 2)$, $X(3, 4)$, $Y(5, 0)$ and $Z(1, -2)$.

(a) Compute the side length WX and both diagonals WY and XZ , to two decimal places. (The other three sides are congruent to WX — you may take that as given.) (b) Give the most specific name for $WXYZ$ and state which implication justifies it. Which single test would have been enough to rule out “rhombus but not square”? (*Plot $WXYZ$ on Grid 6, printed with the directions.*)

Answer: _____